26 A generator produces a current of 60.0 A r.m.s at a voltage of 120 V r.m.s. An ideal transformer steps up the voltage to a peak value of 4500 V and transmits the power through a cable of total resistance 40.0 Ω to the consumers.

What is the percentage of power transmitted to the consumers?

A 94.3 %

B 97.2 %

C 98.6 %

D 99.3 %

Ans: **B**

Primary circuit mean power $\langle P_p \rangle = 120 \times 60.0 = 7200$ W

For the secondary circuit,

$$I_{rms}\left(\frac{4500}{\sqrt{2}}\right) = 7200$$

$$I_{rms} = \left(\frac{7200\sqrt{2}}{4500}\right)$$

Mean power transmitted $P' = 7200 - I_{rms}^{2}R$

$$= 7200 - \left[\left(\frac{7200\sqrt{2}}{4500}\right)^{2} 40.0\right]$$

Therefore, % power transmitted

$$= \frac{P'}{7200} \times 100\%$$

≈ 97.2 %